

UNIT-I ANALYTICAL ASSIGNMENT

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Course

Code: ITA0612

Course: Machine Learning

Slot : A

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Unit-1

Analytical Assignments

1. Explain how Machine Learning can identify poverty-prone regions using socio-economic data.

Ans: Machine Learning can identify regions that are more likely to experience poverty by analysing socio-economic data.

Working:

1. Collect data such as:

- * Household income, Employment rate, Education level, Housing conditions, Access to healthcare.

2. Preprocess the data by removing missing values, errors and duplicate records.

3. Select important features such as income, unemployment, literacy rate, and access to basic facilities.

4. The model predicts the poverty level/risk of different regions.

5. Train an ML model using historical poverty data.

Benefits:

- * Helps government identify areas requiring urgent support.

- * Enables better allocation of welfare funds.

- * Helps target employment, education and healthcare programs.

- * Supports data-driven poverty reduction.

Example: If a district has low income, high unemployment, poor education, and limited healthcare facilities, the ML model may classify it as a high poverty-risk region.

2. Design a Machine Learning model to Identify eligible beneficiaries for government welfare schemes.

Sol: Machine Learning can help governments identify people who are eligible for welfare Schemes based on their socio-economic conditions.

Working:

1. Collect citizen data such as:

- * Annual income, Family size, Employment status, Age, Disability status, Housing condition, Previous benefits received.

2. Clean and preprocess the data.

3. Define the eligibility rules or use historical records of approved applications as training data.

4. Train a model such as, Decision Tree, Random Forest or Logistic Regression.

5. Government officials can verify the prediction before providing benefits.

Benefits:

- * Faster Identification of deserving beneficiaries.

- * Reduces manual work.

- * Reduces errors and duplicate beneficiaries.

- * Helps reach vulnerable people who may be missed.

- * Improves transparency and efficient distribution of resources.

Example:

An ML system can analyse income, family size, employment, and housing conditions to identify families who are likely to qualify for a food subsidy or housing scheme.

Q Explain how Machine Learning can predict crop yield using historical agricultural data.

Ans: Machine Learning can predict the expected crop yield by analysing historical agricultural and environmental data.

Data Used:

- * Previous crop yield, Rainfall, Temperature, Soil quality, Soil moisture, Fertilizer usage, Seed variety.

Working:

1. Collect historical agricultural data.
2. Clean and preprocess the data.
3. Select important factors affecting crop yield.
4. Train an ML model using previous crop data.
5. Algorithms such as Linear Regression, Random Forest or XGBoost can be used.
6. Provide current weather, soil, and farming information to the trained model.
7. The model predicts the expected crop yield.

Benefits:

- * Helps farmers plan production.
- * Helps decide fertilizer and irrigation requirements.
- * Predicts possible low-yield seasons.
- * Helps governments plan food supply.
- * Supports food security and reduces the risk of hunger.

Example:

If historical data shows that low rainfall and high temperature reduce rice production, the model can predict a lower yield for similar future conditions.

4. Develop a Machine Learning solution for crop disease detection using leaf images.

Ans: Machine Learning, especially Deep Learning and Convolutional Neural Networks (CNNs), can detect crop diseases from photographs of plant leaves.

Working:

Step 1 - Image collection

collect images of healthy and diseased leaves.

Step 2 - Image Preprocessing

Resize images, remove noise, and normalize them.

Step 3 - Model Training

Train a CNN using labelled images.

The model learns features such as:

- Spots
- Discoloration
- Leaf damage
- Fungal patterns
- changes in leaf texture

Benefits:

- * Early detection of diseases.
- * Reduces crop losses.
- * Reduces unnecessary pesticide usage.
- * Helps farmers take quick corrective action.
- * Can be implemented through a mobile application.

Example:

A mobile app can allow a farmer to upload a rice leaf image and the CNN can classify it as healthy, bacterial disease, fungal disease or another disease.

Explain how Machine Learning can reduce food wastage in Supermarkets.

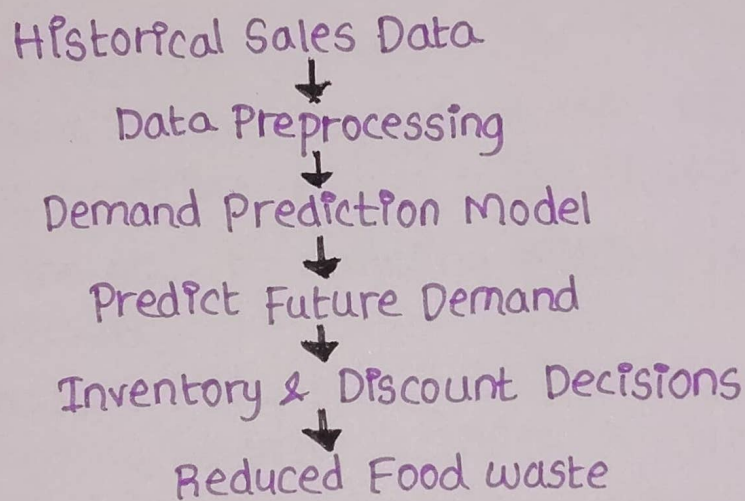
Ans: Machine Learning can help supermarkets predict product demand and expiration risk, allowing them to reduce the amount of food thrown away.

Working:

The Supermarket provides historical data such as:

- * Previous sales, Product demand, Product expiry dates, Weather, Holidays and Festivals, Discounts.

ML Process:



Benefits:

- * Reduces expired food.
- * Improves inventory management.
- * Reduces financial losses.
- * Helps supermarkets provide discounts before expiry.
- * Makes better use of available food.
- * Contributes to food security.